

Shammo Biswas

COMPUTER SCIENCE AND ENGINEERING GRADUATE | DATA SCIENCE AND AI

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Personal Profile

A Computer Science and Engineering graduate with a passion for computer vision, medical imaging, and vision language models, currently pursuing a Masters degree and working as a Research Assistant at the CVIS Lab, BRAC University. My experience spans deep learning research, data analysis, and academic instruction, and my commitment to continuous learning positions me well for contributing to impactful research and innovation.

Education

BRAC University

Masters of Science in Computer Science and Engineering

- CGPA: 4.00 / 4.00

Dhaka, Bangladesh

Oct 2024 - Ongoing

BRAC University

Bachelors of Science in Computer Science and Engineering

- CGPA: 3.76 / 4.00

Dhaka, Bangladesh

Jan 2020 - May 2024

Bangladesh Navy College, Dhaka

Higher-Secondary School Certificate

- Group: Science
- GPA: 5.00 / 5.00

Dhaka, Bangladesh

2017 - 2019

Banophool Adibashi Green Heart College

Secondary School Certificate

- Group: Science
- GPA: 5.00 / 5.00

Dhaka, Bangladesh

2015 - 2017

Work Experience

Department of Computer Science and Engineering, BRAC University

Departmental Research Assistant

- Working under the supervision of Dr. Md. Ashraful Alam at the Computer Vision and Intelligence (CVIS) Lab.
- Research domains: Computer Vision, 3D Vision, Medical Computing, and Vision Language Models.

Dhaka, Bangladesh

Oct 2024 - Ongoing

BRAC University

Instructor

- Conducted a practical session on Machine Learning for Masters of Public Health students at James P Grant School of Public Health.
- Instructed in the CVIS Lab workshop "Training on Development of ML/DL Models" for undergraduate thesis students; delivered sessions on Neural Networks and Video Processing.

Dhaka, Bangladesh

Summer 2025

Department of Mathematics and Natural Science, BRAC University

Student Tutor

- Course Taught: Principles of Physics

Dhaka, Bangladesh

Feb 2024 - May 2024

NETZ, Bangladesh

Data Analyst [Contractual]

- Worked as the data analyst for a Base-line survey on "Promotion of Social Partnership for Empowerment of Marginalized Communities in 6 districts and at national level in Bangladesh (PROSPECT)" in August, 2020.
- Worked as the data analyst for a Base-line survey on "Young people at high schools strengthen women's rights and inclusive governance in Bangladesh (YUKTA)" in January 2022.
- Worked as the data analyst for the End-line survey on "Promotion of Social Partnership for Empowerment of Marginalized Communities in 6 districts and at national level in Bangladesh (PROSPECT)" in July, 2023.

Dhaka

Aug 2020 - Jul 2023

Publications

JOURNAL ARTICLES

A Vision Transformer-Based Hybrid Neural Architecture for Automated Handwritten Bangla Character Recognition and Braille Conversion
Touseef Saleh Bin Ahmed, Tawhidur Rahman, Shammo Biswas, Saifur Rahman Sabuj, Mohammed Belal Bhuiyan, Mohammad Ali Moni, Md Ashraful Alam
Knowledge-Based Systems p. 114546. Elsevier, 2025

OCT-KAN: Advancing Retinal Disease Diagnosis through Adaptive Kolmogorov-Arnold Networks in Optical Coherence Tomography [Under Review]
Md. Fahmid-Ul-Alam Juboraj, Shammo Biswas, Md. Golam Rabiul Alam, Muhammad Iqbal Hossain

CONFERENCE PROCEEDINGS

Enhancing Bangla Video Understanding through Multimodal Feature Integration and Attention-Based Encoder-Decoder Captioning Models for Single-Action Videos [Withdrawn]

Shammo Biswas, Saurav Das, Taimoor Fahim, M.A.B.Siddique Sanjan, Tasnia Alam Tarannum, Md Ashraful Alam, Md Golam Rabiul Alam

Submitted to Asia Conference on Computer and Communications Security (AsiaCCS 2025), 2025, Hanoi, Vietnam

Projects

Enhancing Bangla Video Understanding through Multimodal Feature Integration and Attention-Based Encoder-Decoder Captioning Models for Single-Action Videos

Undergraduate Thesis

- Our thesis focused on generating accurate Bangla captions for single-action videos using a multimodal fusion and attention-based encoder-decoder model. Collected and annotated a new dataset for this purpose.
- Skills Utilized: TensorFlow, Deep Learning, Computer Vision, Natural Language Processing, Attention Mechanisms, Encoder-Decoder Architecture.

Portrait Generation Using GANs: A Deep Learning Approach to Artistic Creation

Research Project

- This project leverages Generative Adversarial Networks (GANs) to generate realistic portraits by training a model with a generator-discriminator architecture, refining output across multiple training epochs.
- Skills Utilized: TensorFlow, GANs, Deep Learning, Computer Vision, Neural Networks, Image Synthesis.

Bladder Tissue Classification Using Vision Transformers (ViT): A Deep Learning Approach for Medical Imaging

Research Project

- This project applies Vision Transformer (ViT) models to classify endoscopic bladder tissue images into different types by utilizing image preprocessing, augmentation, and deep learning techniques.
- Skills Utilized: TensorFlow, Vision Transformers (ViT), Deep Learning, Image Preprocessing, Data Augmentation, Computer Vision.

Skills

Programming Python (Pandas, NumPy, PyTorch, Scikit-learn, TensorFlow), JavaScript (MERN Stack), Java, MySQL, C#, HTML, CSS.

Miscellaneous Microsoft Excel, PowerBi, Linux, Shell (Bash/Zsh), $\text{\LaTeX}$ (Overleaf), Git.

Research Data Collection and Analysis, Experiment Design, Statistical Analysis, Machine Learning and Deep Learning Models, Model Evaluation.

Soft Skills Time Management, Critical Thinking, Collaboration, Clear Communication, Problem-Solving.

Honors and Scholarship

- Appeared 4 times on VC's list and 2 times on Dean's list for outstanding academic results in undergrad.
- Awarded performance-based scholarship in BRAC University.

References

1. Dr. Md. Ashraful Alam

Associate Professor

Department of Computer Science and Engineering, BRAC University

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2. Dr. Md. Golam Rabiul Alam

Professor

Department of Computer Science and Engineering, BRAC University

Email: rabiul.alam@bracu.ac.bd